

NTI Summer Training Program

National Telecommunication Institute (NTI) & ITIDA — Egypt

Source: <https://www.nti.sci.eg/summertraining/>

Program Overview

The Summer Training initiative is a joint program between the National Telecommunication Institute (NTI) and the Information Technology Industry Development Agency (ITIDA), under Egypt's Ministry of Communications and Information Technology (MCIT). It aims to train and qualify Egyptian university students in the pre-graduation stage, building their capabilities in modern, high-value technological fields.

Target students: First, second, and third-year students from faculties of Engineering (Computer & Systems, Communications, and Electronics departments), Computers & Information, Artificial Intelligence, as well as Business Information Systems (BIS) & Business Informatics.

The program is a **free scholarship** for outstanding Egyptian undergraduate students, selected according to each track's conditions. It is delivered both online and in person, at NTI premises and at Digital Egypt Innovation Hubs (Creativa) across Egypt's governorates. The program is part of a national push to train 10,000 students. Graduates receive a certificate accredited by both ITIDA and NTI, along with preparation opportunities for international certifications and hands-on, real-world simulation projects.

Key Facts

Duration	4 weeks
Total training hours	120 hours
Soft / Freelance skills hours	30 hours
Technical training hours	90 hours (3 weeks per technical track)
Cost	Free (scholarship-based, competitive selection)
Delivery mode	Online and in-person (NTI premises & Creativa Digital Egypt Innovation Hubs)
Certificate	Accredited jointly by ITIDA and NTI
Organizers	National Telecommunication Institute (NTI) + ITIDA, under MCIT
Eligibility	1st, 2nd & 3rd year university students (Engineering, Computers & Info, AI, BIS/Business Informatics)

Part 1 — Personal Skills (30 Hours)

A. Soft Skills Track

Scope: Development of non-technical skills essential for personal and professional growth in the digital labor market.

Prerequisite: Basic computer literacy.

Contents (6 modules):

- Communication & Presentation Skills
- Leadership and Teamwork Skills
- Time Management Skills & Business Writing
- Critical Thinking & Problem-Solving Skills
- Negotiation and Conflict Resolution Skills
- Interviewing Skills & CV Writing

B. Freelancing Skills Track

Scope: Equips participants with the skills and knowledge to land freelance jobs, based on best practices in the freelance field; concludes with building a landing page on a freelance platform (e.g. Upwork).

Prerequisite: Basic computer literacy.

Contents:

- Introduction to Freelancing
- Identification of Service and Offering
- Develop a Pricing Structure
- Essentials of Project Management
- Create a Portfolio for Success and Personal Branding
- Delivering a Successful Proposal

Part 2 — Technical Fields (90 Hours, 3 Weeks)

Participants choose one of 8 technical fields. Each field includes several specialized modules (module titles as listed on the official site):

1. Cybersecurity

Builds a progressive path from networking fundamentals to applied security: starts with networking basics (OSI/TCP-IP models, IP addressing & subnetting, switching, routing, VLANs, DHCP/DNS/NAT — broadly aligned with CCNA-style content), then moves into network security concepts (ACLs, firewalls, access control) and hands-on work with Fortinet security appliances.

Modules:

- Network Infrastructure
- Network Security
- Fortinet Cybersecurity

2. Software Development

Covers the web development pipeline from interface design to full production apps: UI/UX design principles and prototyping, front-end web design (HTML/CSS/JS), then two full-stack paths — the MEAN stack (MongoDB, Express.js, Angular, Node.js) and full-stack development using PHP with a relational database.

Modules:

- UI/UX Design
- Web Designer
- MEAN-Stack Web Development

- Full Stack Web Development using PHP

3. Data Science

Starts with Python programming fundamentals, then builds into data analytics and advanced analytics (data wrangling, visualization, statistical/ML techniques), extends into big data analysis tools and workflows, and includes an applied IoT applications module connecting data pipelines to connected devices.

Modules:

- Programming using Python
- Data Analytics
- Advanced Data Analytics
- Big Data Analysis
- Internet of Things (IoT) Applications

4. Cloud Computing

Progresses from cloud computing fundamentals (service models, core cloud concepts) to cloud architecture design (designing scalable, resilient cloud solutions) and datacenter build-out (infrastructure, virtualization, and datacenter operations).

Modules:

- Cloud Essentials
- Cloud Architect
- Building Datacenters

5. Systems Administration

Hands-on system administration on the two major enterprise platforms: Linux administration using Red Hat (users, filesystems, services, shell scripting, networking basics) and Windows Server administration (Active Directory, roles/features, user & resource management).

Modules:

- Linux Red Hat Administration
- Windows Server Administration

6. Artificial Intelligence

Covers core machine learning concepts and algorithms, applying AI to business problems and decision making, computer vision techniques (image processing, object detection/recognition), and NLP (text processing, language models, common NLP tasks).

Modules:

- Machine Learning
- AI For Business
- Computer Vision
- Natural Language Processing (NLP)

7. Electronics

Focuses on embedded systems and digital hardware design: programming and interfacing microcontroller-based systems (sensors, actuators, embedded control), and digital design using FPGAs (HDL-based design and implementation on programmable logic).

Modules:

- Microcontroller-Based Systems
- Digital Design using FPGA

8. Digital Arts

Covers visual and marketing creative skills: graphic design fundamentals and tools, 2D motion graphics and animation, and digital marketing (content, campaigns, and platforms for reaching audiences online).

Modules:

- Graphic Design
- 2D Motion Graphics
- Digital Marketing

Note: the module-by-module descriptions above are general explanations of what each named module typically covers (based on standard curricula and past participants' accounts), since the official site loads each module's exact scope/objectives text through JavaScript pop-ups that aren't accessible as static text. The module **titles** and the field/track structure themselves are taken directly from the official NTI page.

Registration

Registration is done through an NTI account (profile): sign in (or create an account if you don't have one), then go to the "Summer Internships Program Modules" section in your profile and press "Add Module" to select your chosen module/track. A separate "Private Summer Training Course" path exists for students who receive a student code from the NTI financial team.

Official page: <https://www.nti.sci.eg/summertraining/>

Registration page: <https://www.nti.sci.eg/summertraining/register.php>

Modules details page: <https://www.nti.sci.eg/summertraining/summermodules.php>

Compiled from the official NTI Summer Training website and related public announcements (NTI, ITIDA, EgyptInnovate). For exact dates, deadlines and eligibility confirmation, check the official site directly, as these can change each cycle.